OPERATIONS RUNBOOK

Proxmox VE — Ceph Manager (MGR) Configuration Guide

Setting a Specific Node as Active Ceph Manager • Starting Standby MGR Daemons • Future Reference SOP

Cluster: cisai-pve
Nodes: prox22 / prox23 / prox24
Goal: prox22 = Active Ceph MGR

VERIFIED & APPLIED 2026-09-19

1. Background & Problem Statement

The Ceph Manager (`ceph-mgr`) daemon provides monitoring, orchestration, and dashboard services for the Ceph distributed storage cluster. Only **one node is active** at a time; the rest run as standbys. By default, whichever node starts first wins the election.

⚠ Problem Observed

Only `prox24` had Ceph MGR running (shown as ✓ in Proxmox UI). `prox22` and `prox23` showed **?** because their `ceph-mgr` daemons were not started. The goal was to make `prox22` the active manager (consistent with it being CRM Master).

✓ Result Achieved

`prox22` is now the **Active Ceph Manager**. All 3 nodes show ✓ in the Proxmox UI Ceph → Manager view. `prox22` is now master for both Corosync CRM and Ceph MGR.

2. Complete Step-by-Step Commands Executed

1 Check current Ceph Manager status (run from any node)

SSH into any cluster node and check which node holds the active MGR role:

```
# SSH into any node (example: prox23)
ssh root@10.31.6.173

# Check current active Ceph Manager
ceph mgr stat

# Example output (prox24 was active, others offline):
{
    "epoch": 31,
    "available": true,
    "active_name": "prox24",
    "num_standby": 0
}
```

Note: `num_standby: 0` means only prox24 had ceph-mgr running. prox22 and prox23 daemons were inactive.

2 Start ceph-mgr daemon on prox22 (SSH into prox22)

```
# SSH into prox22
ssh root@10.31.6.172

# Enable ceph-mgr to start on boot and start it now
systemctl enable ceph-mgr@prox22
systemctl start  ceph-mgr@prox22

# Verify it is running
systemctl is-active ceph-mgr@prox22
active
```

3 Start ceph-mgr daemon on prox23 (SSH into prox23)

```
# SSH into prox23
ssh root@10.31.6.173

# Enable and start ceph-mgr on prox23
systemctl enable ceph-mgr@prox23
systemctl start  ceph-mgr@prox23

# Verify
systemctl is-active ceph-mgr@prox23
active
```

4 **Wait ~10 seconds, then confirm all 3 are registered as standbys**

```
# Wait for standbys to register, then re-check from any node
sleep 10 && ceph mgr stat

# Expected output — all 3 registered (num_standby: 2):
{
    "epoch": 38,
    "available": true,
    "active_name": "prox24",
    "num_standby": 2
}
```

num_standby: 2 confirms prox22 and prox23 are now registered and ready to take over as active manager.

OPERATIONS RUNBOOK

Ceph MGR Failover Commands & Quick Reference

Forcing Active Manager Election • Verification Commands • Future SOP

Cluster: cisai-pve
Active Node: prox22 (target)
100% VERIFIED

5 Force prox24 to step down (run from prox22 or any node)

The `ceph mgr fail <name>` command forces a manager to stop being active. Ceph then holds a new election among the remaining standbys.

```
# Force the current active manager (prox24) to step down
ceph mgr fail prox24

# Wait ~5-8 seconds for new election to complete
sleep 6 && ceph mgr stat
```

Note: The election is randomized. prox22 or prox23 may win. If prox23 wins, proceed to the next step.

6 If prox23 won instead — fail it too to push prox22 to win

```
# If prox23 became active, fail it as well
ceph mgr fail prox23

# Wait for election
sleep 8 && ceph mgr stat

# prox22 should now be the only available candidate and win:
{
    "epoch": 49,
    "available": true,
    "active_name": "prox22",
    "num_standby": 1
}
```

✓ **Success!** `active_name: "prox22"` confirms prox22 is now the active Ceph Manager.

3. One-Shot Script (Future Use) — Run from Any Node

Copy and run this script on any cluster node to automatically elect prox22 as active Ceph Manager:

```
#!/bin/bash
# Script: set_ceph_mgr_prox22.sh
# Purpose: Force prox22 to become the active Ceph Manager
# Run from: any Proxmox node (prox22, prox23, or prox24)

# Step 1: Ensure ceph-mgr is running on prox22 and prox23
ssh root@10.31.6.172 "systemctl enable --now ceph-mgr@prox22"
ssh root@10.31.6.173 "systemctl enable --now ceph-mgr@prox23"
sleep 10

# Step 2: Fail active managers until prox22 wins
for i in 1 2 3; do
  ACTIVE=$(ceph mgr stat | python3 -c "import sys,json; print(json.load(sys.stdin).get('active_name',''))")
  [ "$ACTIVE" = "prox22" ] && echo "SUCCESS: prox22 is active!" && break
  echo "Failing $ACTIVE ..." && ceph mgr fail "$ACTIVE" && sleep 8
done

# Step 3: Final verification
ceph mgr stat
```

4. Verification & Quick Reference Commands

Purpose	Command	Expected Output
Check active MGR	`ceph mgr stat`	`active_name: "prox22"`
List all MGR daemons	`ceph mgr dump`	Shows active + all standbys

Purpose	Command	Expected Output
Check MGR service on a node	`systemctl status ceph-mgr@prox22`	`active (running)`
Force MGR failover	`ceph mgr fail <node-name>`	New election triggered
Enable MGR at boot	`systemctl enable ceph-mgr@<node>`	Enabled in systemd
Start MGR daemon	`systemctl start ceph-mgr@<node>`	Daemon starts as standby
Cluster overall health	`ceph -s`	`HEALTH_OK`

5. Node IP Reference

Hostname	Management IP	Storage Fabric IP	Roles
prox22	`10.31.6.172`	`192.168.10.22`	CRM Master + **Active Ceph MGR**
prox23	`10.31.6.173`	`192.168.10.23`	Standby CRM + Standby MGR
prox24	`10.31.6.174`	`192.168.10.24`	Workload Host + Standby MGR

Why ceph-mgr was inactive on prox22/prox23: The `ceph-mgr@<node>` systemd service must be explicitly enabled and started on each node. It does not auto-start after Proxmox VE installation unless configured during initial Ceph deployment. The `systemctl enable --now` command both enables it for future reboots and starts it immediately.